Lecture-2
Introduction to Fluid Mechanics

Saikat Mukherjee

Department of Mechanical Engineering, IIT Kharagpur

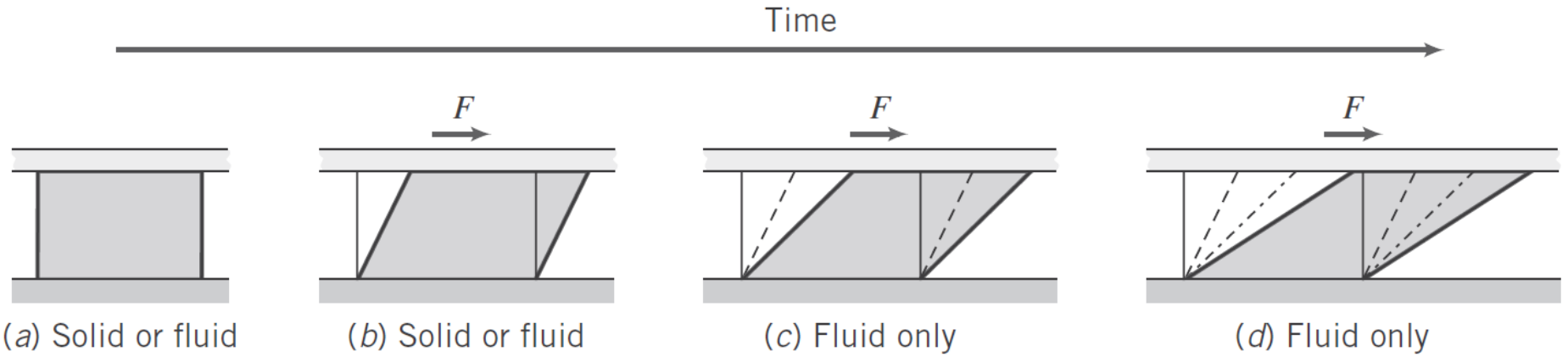
Learning Objectives

By the end of this lecture, you should be able to

1. Differentiate **solid versus fluid** behavior
2. Evaluate the **validity of the continuum hypothesis** for a given engineering problem
3. Recognize **some common variables** used to describe fluid properties
4. Deconstruct a **stress tensor to identify** the normal and shear stress components acting on a fluid element

What is a fluid?

- A fluid is a substance that **deforms continuously** under the application of a **shear stress**

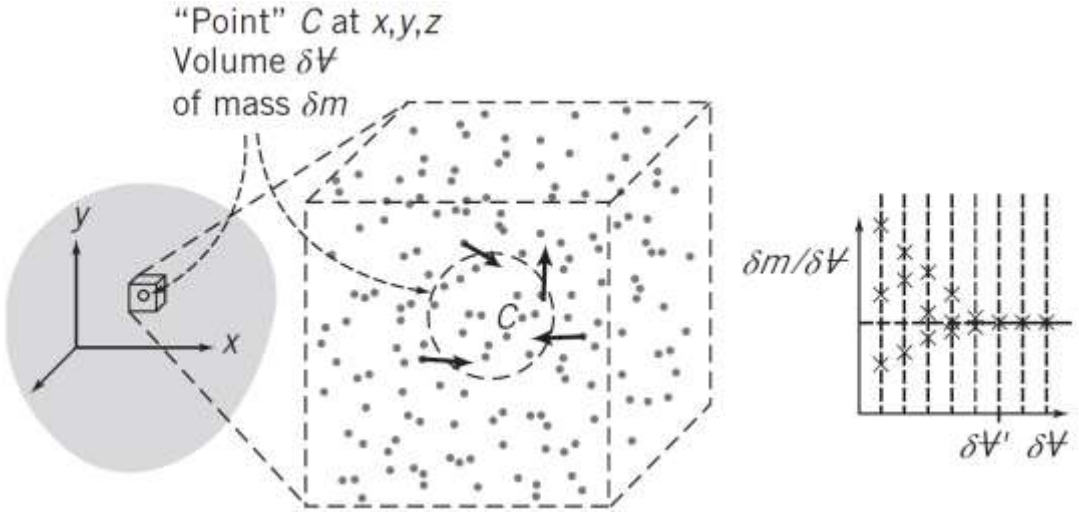


- The property of a fluid that **quantifies the resistance force** acting on fluids when there is relative motion between fluid parcels is called **viscosity.**

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Fluid as a Continuum

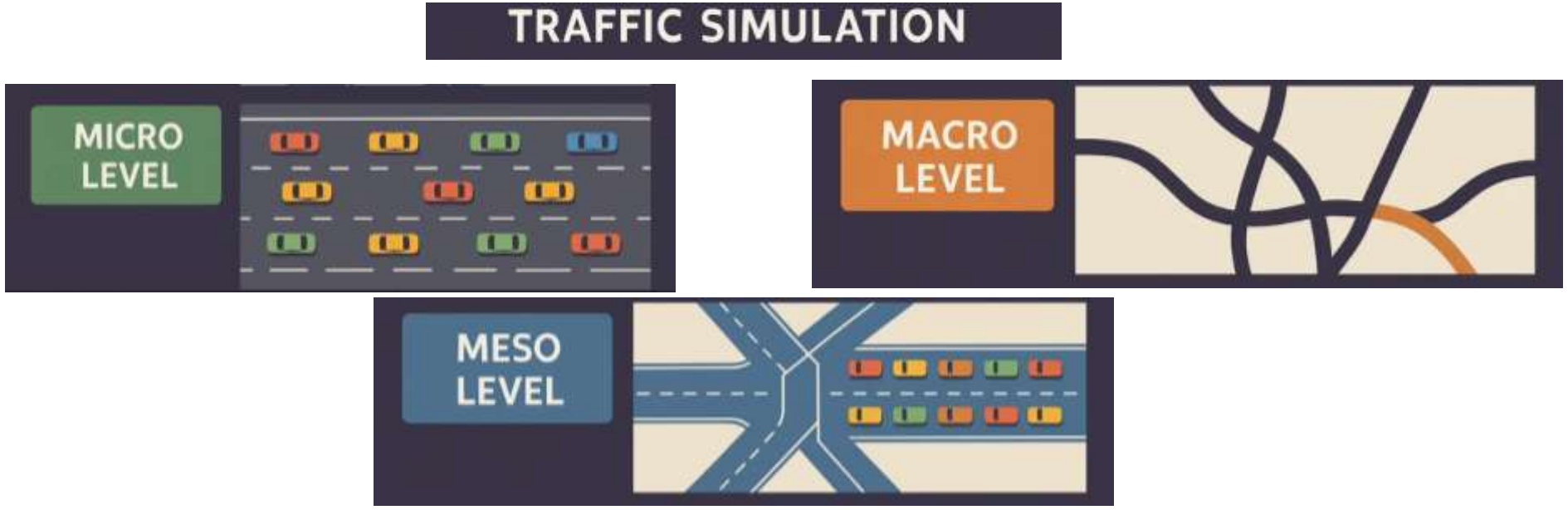
- The **continuum assumption** implies that physical properties are imagined to be **distributed throughout space**.
- Events in the microscopic world are not governed by continuum laws.
- As a consequence of the continuum assumption, fluid properties can be considered continuous functions of space and time.



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When is the continuum assumption valid?

- **Knudsen number** is the ratio between the molecular mean free path and a representative physical length scale
- If the **Knudsen number is comparable to 1**, the continuum assumption is no longer a good approximation.



https://medium.com/ak-modeling-engineering/the-art-and-science-of-traffic-modeling-673627d3788f

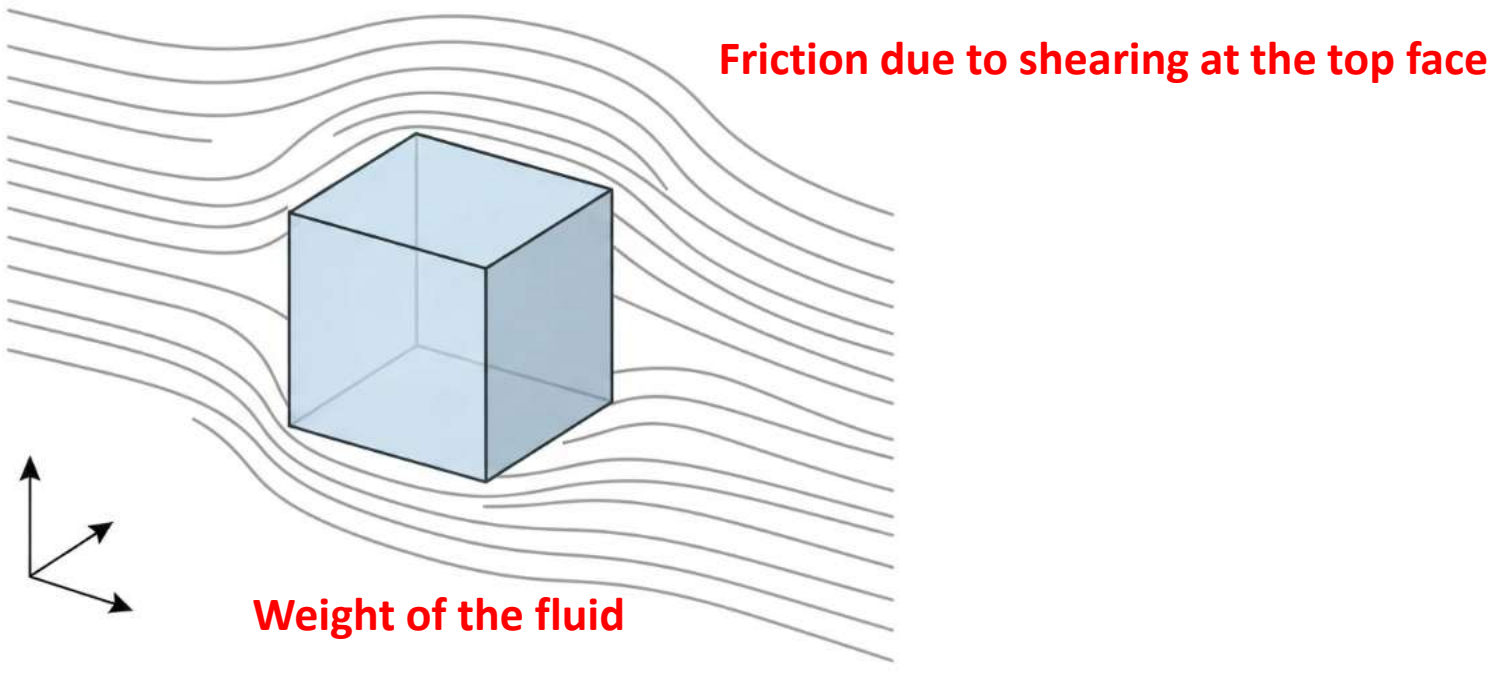
Common physical properties of fluids

- **Density** is defined as the mass of a fluid per unit volume.
- **Specific Gravity** is another way to represent density.
- The **velocity** of a fluid can be interpreted as the velocity of a fluid particle passing through the same point at the same instance of time
- When fluid properties do not change over time, the **flow field is called steady**.

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Forces on fluids

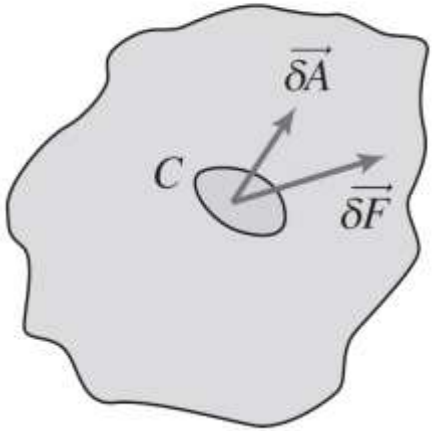
- The forces experienced by a fluid particle can be classified into surface and body forces.
- **Surface forces** are generated on interaction with other fluid particles.
- **Body forces** are experienced throughout the fluid particle

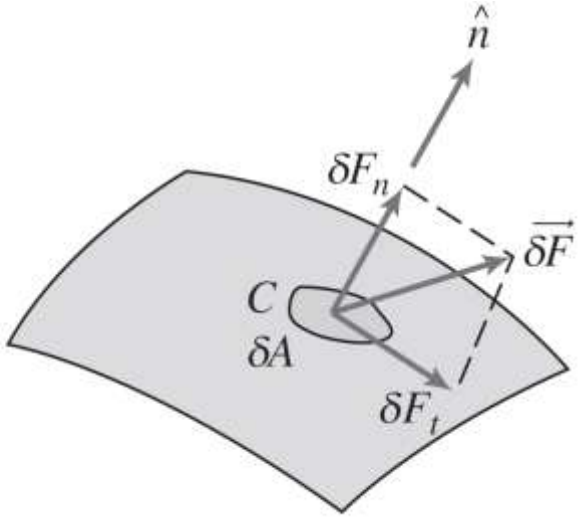


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Types of stresses in a fluid

- **Surface forces** on a fluid particle **lead to stresses**.
- The stresses can be classified into **normal and shear stresses**





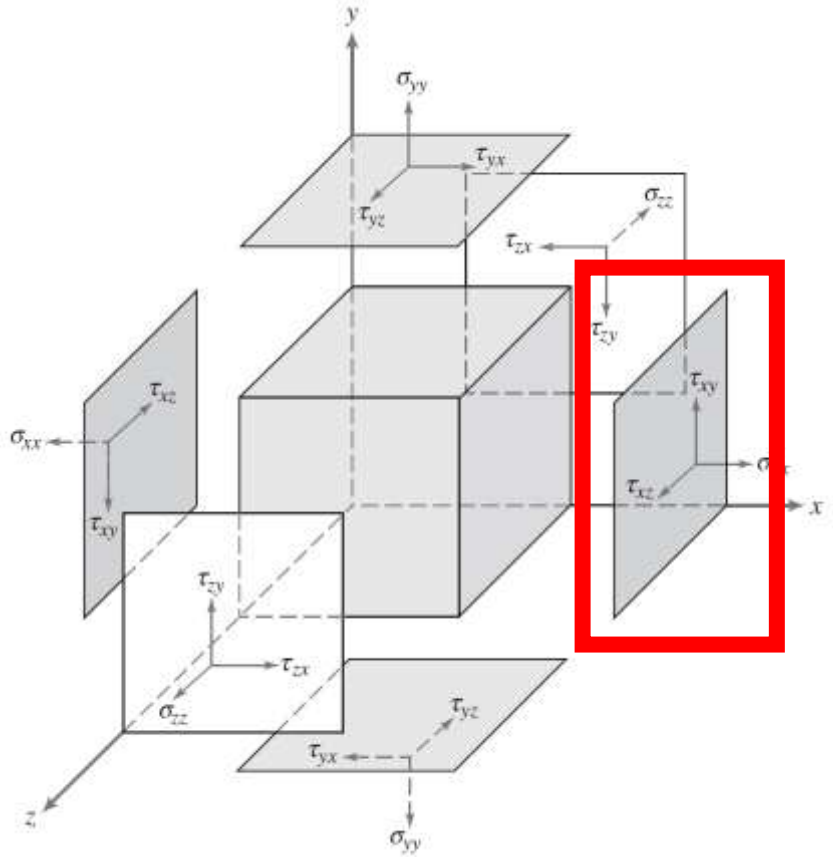
$$\sigma_n = \lim_{\delta A_n \to 0} \frac{\delta F_n}{\delta A_n}$$

$$\tau_n = \lim_{\delta A_n \to 0} \frac{\delta F_t}{\delta A_n}$$

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State of stress of a fluid

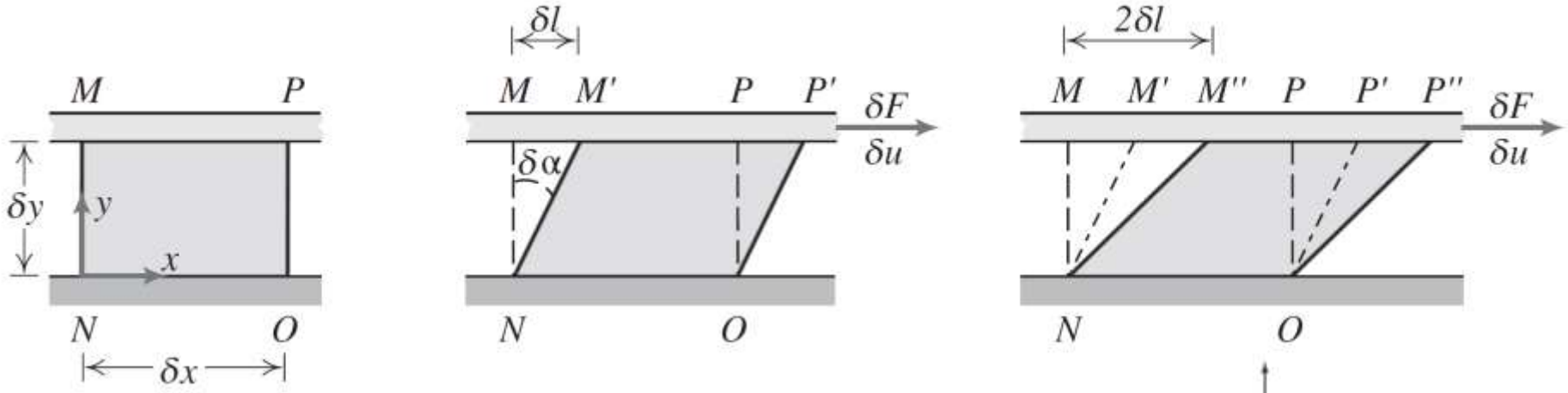
- In an orthogonal coordinate system, the components of the force and area elements can be used to define the **state of stress**



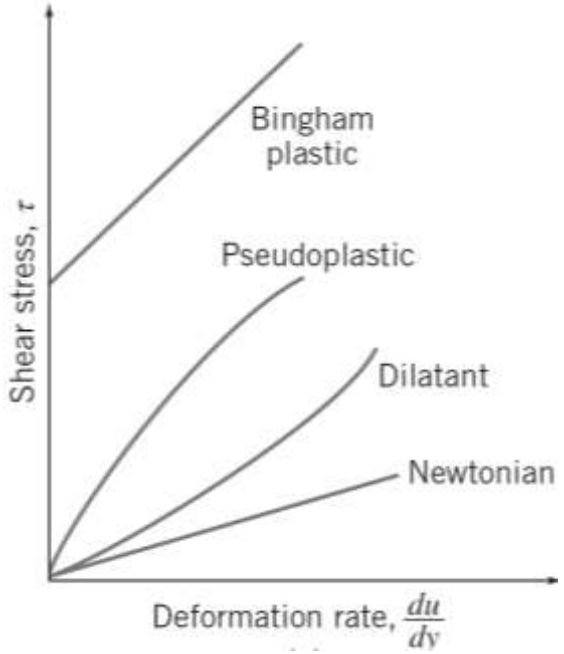
$$\sigma_{xx} = \lim_{\delta A_x \to 0} \frac{\delta F_x}{\delta A_x}$$

$$\tau_{xy} = \lim_{\delta A_x \to 0} \frac{\delta F_y}{\delta A_x} \quad \tau_{xz} = \lim_{\delta A_x \to 0} \frac{\delta F_z}{\delta A_x}$$

Another property of a fluid -Viscosity



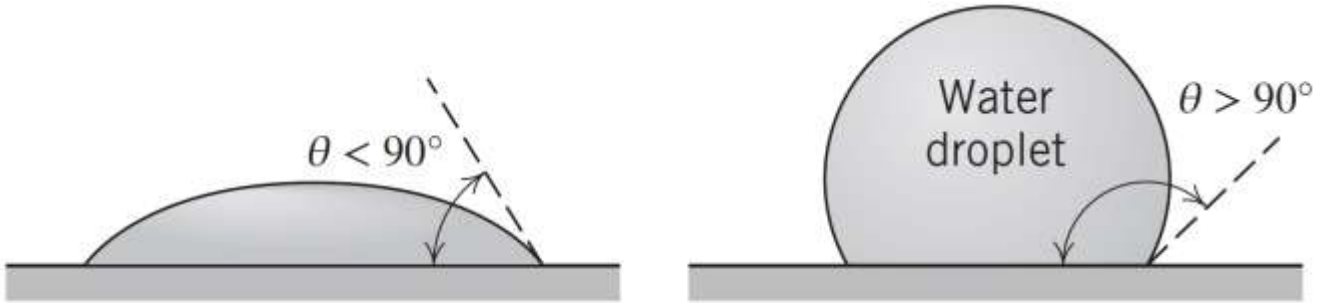
- Based on the relation between the shear stress and strain rate, we can classify fluids into **Newtonian and Non-Newtonian fluids.**
- The constant of proportionality, which represents the ratio of the shear stress to the strain rate for a Newtonian fluid, is the **dynamic viscosity**.



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Other common properties of fluids

- Surface tension is the elastic tendency of a liquid's surface that makes it acquire the **least surface area** possible.



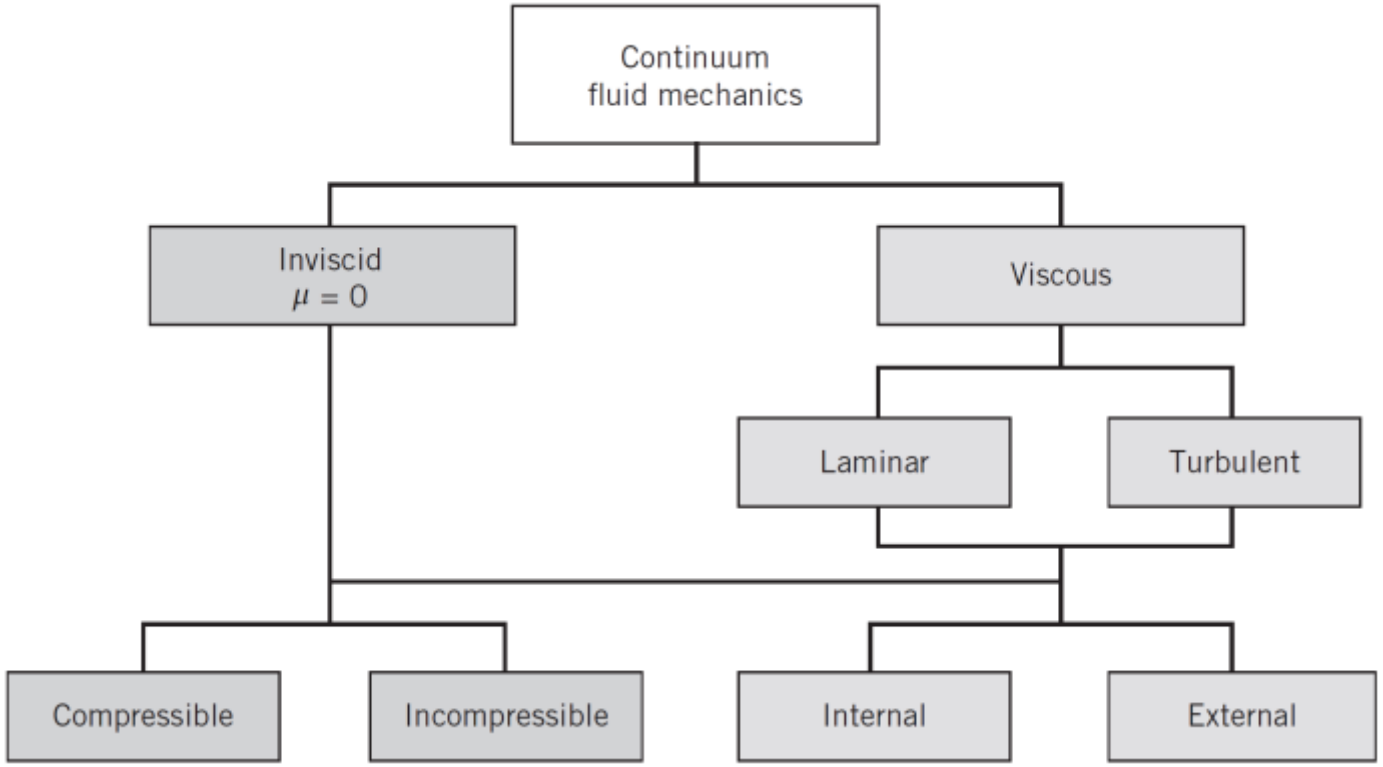
(*a*) A "wetted" surface (*b*) A nonwetted surface

- The bulk modulus of a fluid measures its **resistance to compression**.

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Loose classification of fluid flows

- We can classify fluid flows into various categories loosely as follows, and we will revisit each of these concepts in this course



Revisiting the learning objectives

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